PHYSICAL DI PROPOSAL / NOT DOWNLOADED		
Proposed DI	Function	Logical mapping
P00	Safety auxiliary	→ M01000
P01	START NO	→ M01001
P02	STOP NC healthy1	→ M01002=NOT input
P03	Process RESET NO	→ M01003
P04 /P05 /P06	Infeed /test /outfeed	→ M01004/05/06
P07	Door closed diagnostic	→ M01007
P08	Clamp closed	→ M01008
P09 /P0A /P0B	PASS /FAIL /RETEST arrived	→ M01009/0A/0B
P0C..P0F	Spare /unassigned	Not M0100C environment

RELAY DO / 4 POINTS PER COM		
COM group	Proposed DO	Loads
COM0 /safe24V	P20 /P21 /P22 /P23	Clamp /PASS /FAIL /RETEST
COM1 /safe24V	P24 /P25..P27	Forward run relay /reserved
COM2 /control24V	P28 /P29 /P2A /P2B	Green /yellow /red /buzzer
COM3	P2C..P2F	Spare; supply selection later

WHY THIS PHYSICAL PROPOSAL DIFFERS

Avoid a COM bank mixing safe actuators and maintained alarm indicators.
LS M01102 clamp → proposed P20; M01103/04/05 gates → P21/22/23.
LS M01100 forward → proposed P24 relay; indicators M01106..09 → P28..2B.
Actual terminal labels and ratings require the exact PLC manual.

INTERFACE EXAMPLES / FINAL COMPONENTS STILL REQUIRED

DI example: PNP24V sensor, brown +24V, blue0V, signal PLC input, input COM0V. Choose actual sensor polarity before wiring.
Relay DO example: external protected24V output COM dry contact load 0V. Use an interposing relay/suppression where required.
DC coil suppression is connected across the coil with correct polarity; PLC relay outputs are not transistor sources and do not create24V.
The physical mapping is a proposal, not the current LS program or an automatic address–number copy. IO_MAP/P_OUTPUT are required at real–hardware commissioning.

Official XBC–DR32H:16 DI,16 relay DO;4 outputs/COM. Contact and bank ratings depend on load conditions; motor power never passes through PLC output contacts.

NO.	DATE	REVISION
R0	2026-10-22	CONCEPT / NOT FOR CONSTRUCTION

DWG. TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

PLANNED PHYSICAL DI AND RELAY DO MAPPING

DRAWN BY	DATE
J. H. KO	2026-10-14

DWG. NO
020

SHEET NO
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